

State of Developer Ecosystem 2024

Language Trends, Satisfaction & Strategic Shifts

23,262

Developers Surveyed

8 Years

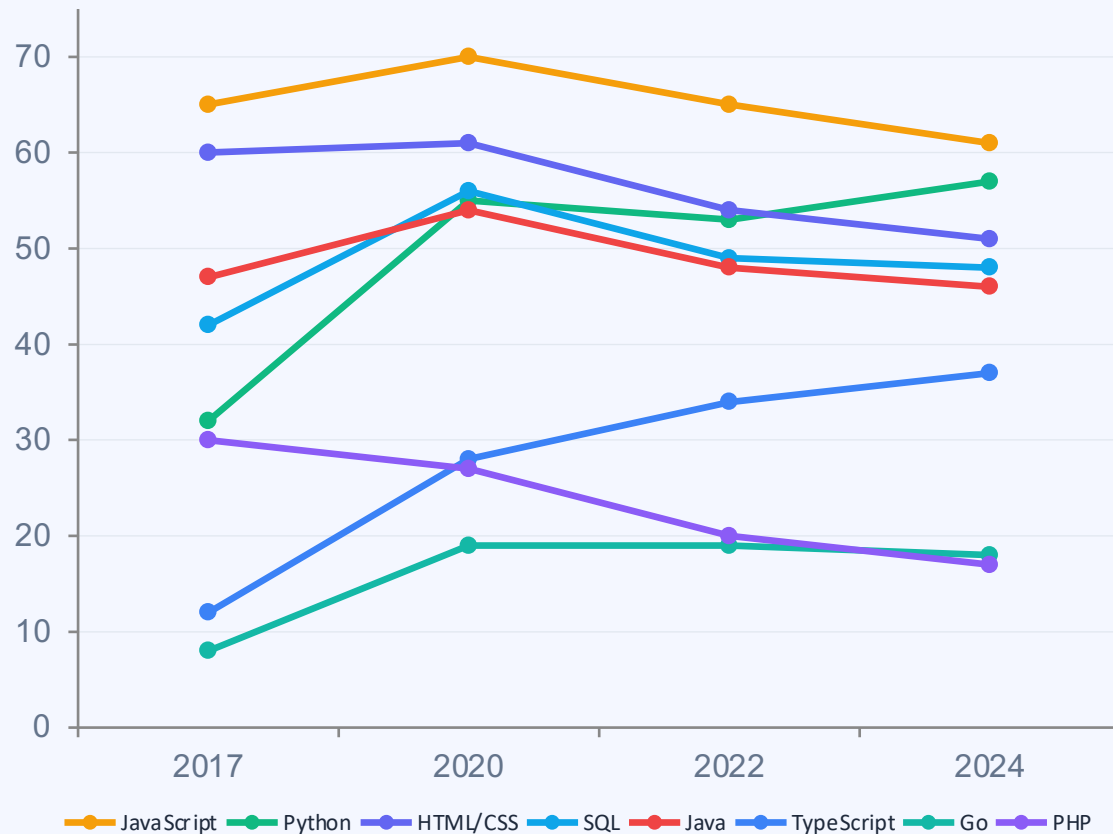
Longitudinal Data

20+

Languages Tracked

Annual survey of 23,262 developers worldwide, covering languages, tools, AI attitudes, and career trends

Python Closes the Gap: 8 Years of Language Usage Shifts



Key Shifts

▲ Python

32% → 57%

▲ TypeScript

12% → 37% (3×)

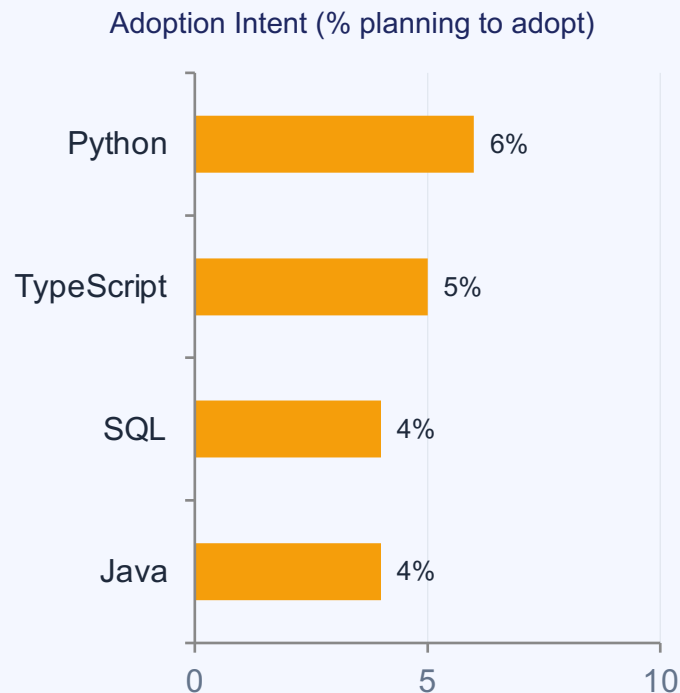
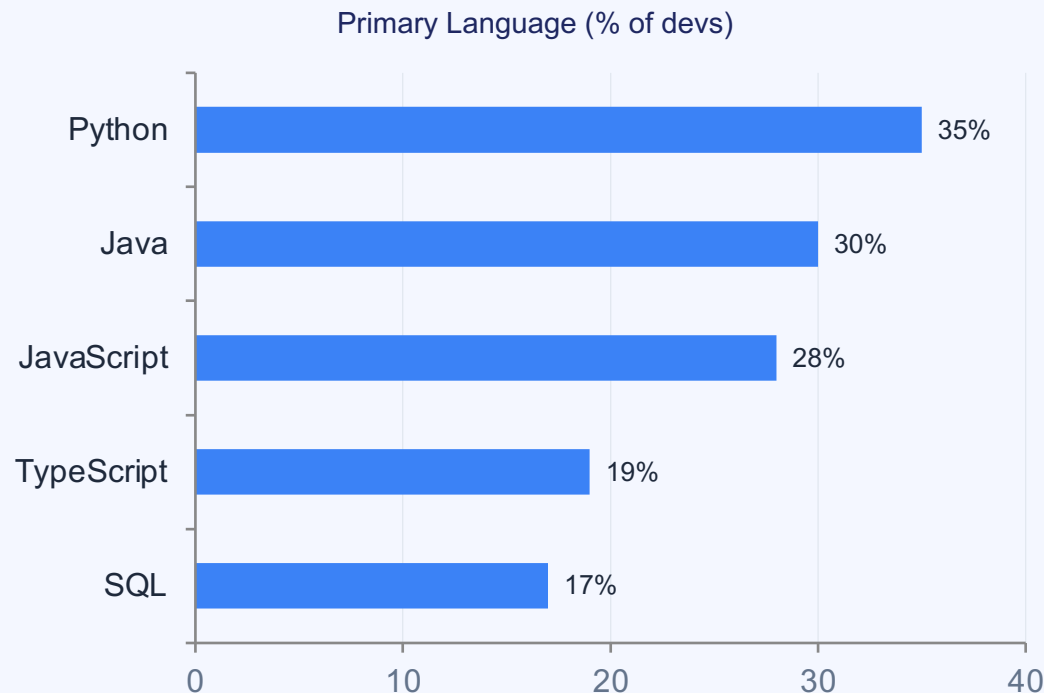
▼ JavaScript

70% peak → 61%

▼ PHP

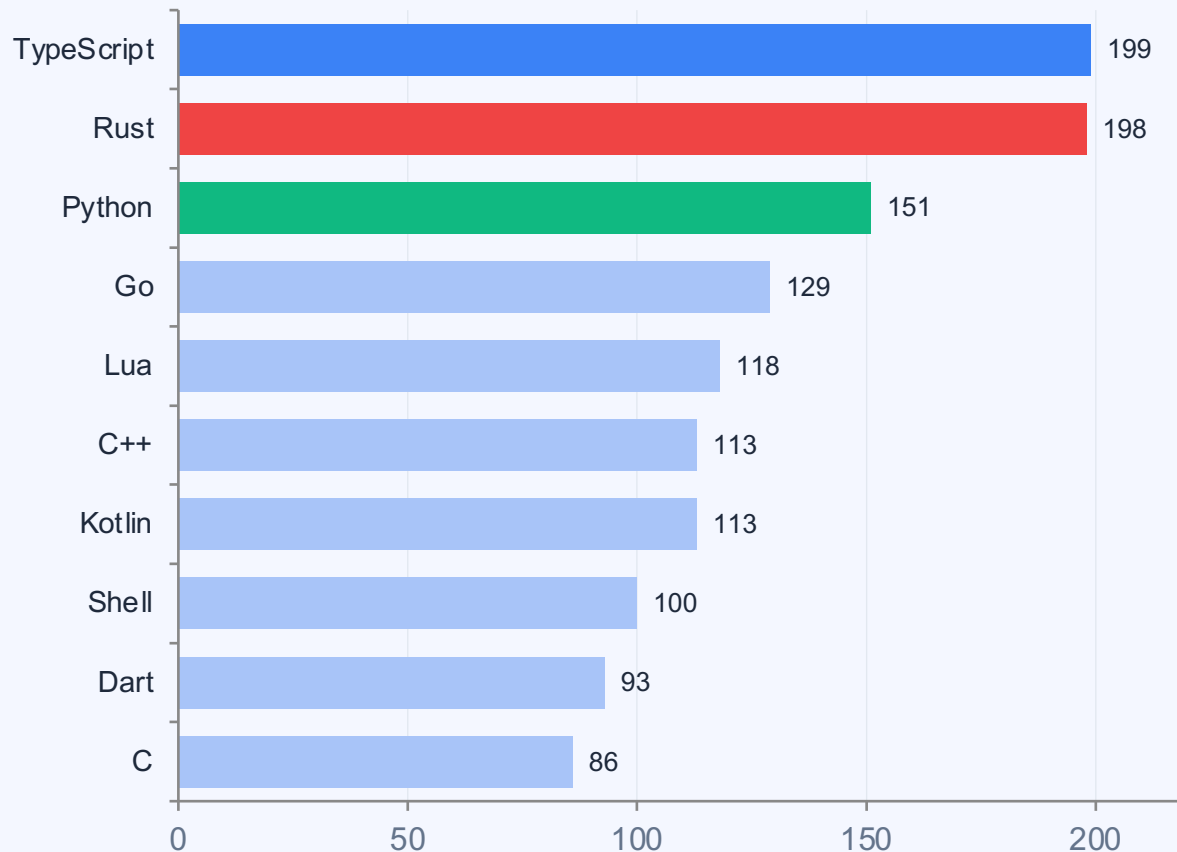
30% → 17%

Python Is Now the #1 Primary Language at 35%



Ecosystem lock-in is strong — adoption intent stays below 6% across all languages, confirming developers rarely switch primary stacks.

TypeScript and Rust Top Developer Satisfaction, Nearly Tied at 199 vs 198



Top 3 Most Loved

TypeScript 199

Type safety + modern DX wins developers over

Rust 198

Memory safety without GC — nearly tied with TS

Python 151

Beloved for productivity & AI/ML dominance

Scores reflect net developer satisfaction index

Each Language Owns a Domain: AI Belongs to Python, UI to TypeScript

Domain	Top Language	Share	Runner-Up
AI / Machine Learning	Python	31%	R, Julia
Data Science & Analytics	Python	28%	R, SQL
Data Processing	Python	17%	SQL, Java
User Interface	TypeScript	62%	JavaScript (58%)
APIs & Services	Java	49%	Python, Go
Application Logic & Workflows	Java	55%	Python, C#
System Tools & Components	C++	27%	C, Rust
Integrating with APIs	Python	38%	JavaScript

What This Means for Your Technology Strategy in 2025

01 Invest in Python for AI/ML Teams

Python is undisputed: 31% of AI/ML, 28% data science, 57% overall. Safest AI-team hire.

02 TypeScript Is the Future of Full-Stack

62% UI share, 37% overall, satisfaction 199. TS investments compound across frontend and backend.

03 Rust & Go: Infrastructure Growth Plays

Go at 18%; Rust still climbing. Both deliver compiler safety for microservices & cloud-native tooling.

04 Shrinking PHP & Ruby Hiring Pools

PHP 30% → 17%; Ruby 10% → 4%. Plan succession paths and modernization roadmaps now.

05 Type Safety Correlates with Retention

TypeScript 199, Rust 198 — both prioritize type systems. Factor this into your DX investment strategy.